

Total No. of Questions : 4]

SEAT No. :

PD105

[Total No. of Pages : 1

[6410]-426

T.E. (Artificial Intelligence and Data Science) (Insem)

ARTIFICIAL NEURAL NETWORK

(2019 Pattern) (Semester - II) (317531)

Time : 1 Hour]

[Max. Marks : 30

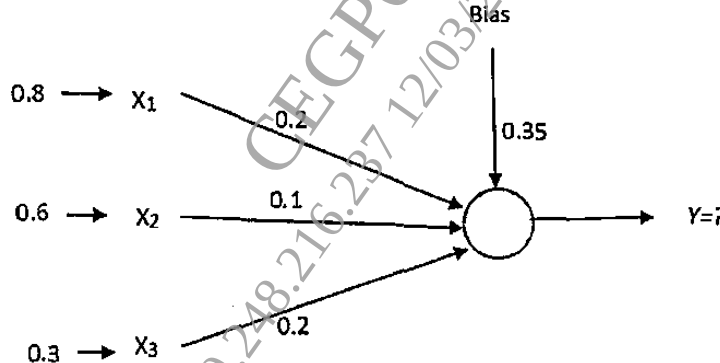
Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary

- Q1)** a) What is neural network topology? Explain any three topologies in detail. [6]
b) Explain briefly McCulloch Pitt's (MP) artificial neuron model. Give its limitations. [5]
c) Distinguish between Biological Neural Network and Artificial Neural Networks. [4]

OR

- Q2)** a) Discuss briefly the structure and function of a biological neuron. [7]
b) Obtain the output of the neuron Y for the network shown in figure using activation function as Binary Sigmoidal and Bipolar Sigmoidal. [8]



- Q3)** a) Write and explain Hebbian learning Algorithm. [6]
b) What is error correction learning? Explain in detail with diagram. [5]
c) Differentiate between Feed Forward and Feedback neural network. [4]

OR

- Q4)** a) Draw the architecture of multilayer feed forward networks. Explain input layer, hidden layer & output layer computations in multilayer feed forward networks. [7]
b) Explain perceptron learning algorithm and implement OR function using Perceptron network. [8]

